# Naveen Jindal School of Management

## Computer Information Systems and Technology (BS)

## Bachelor of Science in Computer Information Systems and Technology

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

## Faculty

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**Professors:** Ashiq Ali, Alain Bensoussan, Gary Bolton, Metín Çakanyildirim, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Dorothée Honhon, Kyle Hyndman, Varghese S. Jacob, Sanjay Jain, Ganesh Janakiraman, Elena Katok, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Zhiang (John) Lin, Sumit K. Majumdar, Stanimir Markov, Amit Mehra, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Naim Bugra Ozel, Özalp Özer, Mike W. Peng, Hasan Pirkul, Cuili Qian, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Suresh P. Sethi, Kathryn E. Stecke, Riki Takeuchi, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

**Associate Professors:** Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Michael Hasler, Bin Hu, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Maria Loumioti, Lívia Markóczy, Ramachandran (Ram) Natarajan, H. Dennis Park, Anyan Qi, Young U. Ryu, Serdar Simsek, Harpreet Singh, Upender Subramanian, Shouqiang Wang, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

**Assistant Professors:** Khai Chiong, Rafael Copat, Soraya Fatehi, Andrew Frazelle, Ying Huang, Joonhwi Joo, Sora Jun, Jason Kautz, Tongil Kim, Sheen Levine, Christopher Mace, Samir Mamadehussene, Jean-Marie Meier, Zixuan Meng, Radha Mookerjee, Jedson Pinto, Ignacio Rios Uribe, Alejandro Rivera Mesias, Simon Siegenthaler, Kirti Sinha, Shujing Sun, Xiaoxiao Tang, Shervin Tehrani, Ashwin Venkataraman, Guihua Wang, Hongchang Wang, Pingle Wang, Junfeng Wu, Steven Xiao

**Professor Emeritus:** R. Chandrasekaran

**Associate Professor Emeritus:** David J. Springate

**Clinical Professors:** John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Paul Convery, Tevfik Dalgic, Howard Dover, Randall S. Guttery, William Hefley, Robert Hicks, Robert Kaiser, Marilyn Kaplan, Van Latham, Sonia Leach, Peter Lewin, Jeffrey Manzi, John McCracken, Larry Norton, Divakar Rajamani, Daniel Rajaratnam, Kannan Ramanathan, Prakash Shrivastava, Mark Thouin, McClain Watson, Jeff Weekley, Habte Woldu, Fang Wu

**Clinical Associate Professors:** Shawn Alborz, Dawn Owens, Carolyn Reichert, Avanti P. Sethi, Ramesh Subramoniam, Aysegul Toptal, David Widdifield

**Clinical Assistant Professors:** Athena Alimirzaei, Christina (Krysta) Betanzos, Moran Blueshtein, Jeffery (Jeff) Hicks, Dupinderjeet Kaur, Revansiddha Khanapure, Liping Ma, Parneet Pahwa, Jason Parker

**Professors of Instruction:** Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Suzette Plaisance Bryan, Matt Polze, Luell (Lou) Thompson

**Associate Professors of Instruction:** Judd Bradbury, Monica E. Brussolo, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Kathryn Lookadoo, Sarah Moore, Hirofumi Nishi, Daniel Sibley, Agnieszka Skuza, Hubert Zydorek

**Assistant Professors of Instruction:** Negin Enayaty Ahangar, Daniel Karnuta, Joseph Mauriello, Rasoul Ramezani, Gaurav Shekhar

**Professors of Practice:** Gregory Ballew, Tiffany A. Bortz, Ranavir Bose, Alexander Edsel, Rajiv Shah, Donald Taylor, Keith Thurgood

**Associate Professors of Practice:** Nozar Hassanzadeh, Jackie Kimzey, Julie Lynch, David Parks, Margaret Smallwood, Steven Solcher, Kathy Zolton

**Assistant Professors of Practice:** Khatereh Ahadi, Steven Haynes, Abu Naser Islam, Edward Meda, Paul Nichols, Timothy Stephens, Guido Tirone, Robert Wright

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

**Communication: 6 semester credit hours**

Select any 6 semester credit hours from Communication Core courses

**Mathematics: 3 semester credit hours**

MATH 1325 Applied Calculus I[1, 2]

Or select any 3 semester credit hours from Mathematics Core courses

**Life and Physical Sciences: 6 semester credit hours**

Select any 6 semester credit hours from Life and Physical Sciences Core courses

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from Creative Arts Core courses

**American History: 6 semester credit hours**

Select any 6 semester credit hours from American History Core courses

**Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from Government/Political Science Option courses

**Social and Behavioral Sciences: 3 semester credit hours**

Choose one of the following:

BA 1310 Making Choices in Free Market Systems[1, 2]

BA 1320 Business in a Global World[1, 2]

ECON 2301 Principles of Macroeconomics[1, 2]

ECON 2302 Principles of Microeconomics[1, 2]

Or select any 3 semester credit hours from Social and Behavioral Sciences Core courses

**Component Area Option: 6 semester credit hours**

Choose one of the following:

BA 1310 Making Choices in Free Market Systems[1, 2]

BA 1320 Business in a Global World[1, 2]

ECON 2301 Principles of Macroeconomics[1, 2]

ECON 2302 Principles of Microeconomics[1, 2]

And select 3 semester credit hours from Component Area Option Core

Or select any 6 semester credit hours from Component Area Option Core courses

II. Major Requirements: 76 semester credit hours

**Major Preparatory Courses: 21 semester credit hours beyond Core Curriculum**

ACCT 2301 Introductory Financial Accounting[1]

ACCT 2302 Introductory Management Accounting[1]

BLAW 2301 Business and Public Law[1]

ITSS 3311 Introduction to Programming[3]

ITSS 4381 Object Oriented Programming with Python

or ITSS 3312 Object-Oriented Programming

OPRE 3333 Quantitative Business Analysis[4]

OPRE 3360 Managerial Methods in Decision Making Under Uncertainty[4]

Complete the following courses if not already completed as part of the Core Curriculum Requirements

MATH 1325 Applied Calculus I[1, 2] (Required for all JSOM undergraduate students. May elect to substitute MATH 2413 or MATH 2417 for MATH 1325.)

Choose two of the following if not completed in Core Curriculum:

BA 1310 Making Choices in Free Market Systems[1, 2]

BA 1320 Business in a Global World[1, 2]

ECON 2301 Principles of Macroeconomics[1, 2]

ECON 2302 Principles of Microeconomics[1, 2]

**Major Core Courses: 28 semester credit hours**

BA 1105 Professional Development (JSOM first-time-in-college freshmen are required to take BA 1105 in their first semester) or BA 3105 Professional Development (Transfer students and students new to JSOM are required to take BA 3105 in their first semester)

BCOM 3300 Business Communication

BUAN 3301 AI in Business

FIN 3320 Business Finance

IMS 3310 International Business

ITSS 3300 Information Technology for Business

MKT 3300 Principles of Marketing

OBHR 3310 Organizational Behavior

or OBHR 3330 Human Resource Management in Business

OPRE 3310 Supply Chain and Operations Management

**Experiential Learning**

A capstone course is required for all undergraduate JSOM students

ITSS 4395 Capstone Senior Project - Information Systems

or BPS 4395 Capstone Senior Project - Business[5]

A practicum experience of at least 160 working hours is required, with registration in the following course.

ITSS 4090 Information Technology and Systems Internship[6]

Students are required to complete 100 hours of community engagement with registration in the course below.

BA 4095 Social Sector Engagement and Community Outreach Practicum[7]

**Major Related Courses: 18 semester credit hours**

ITSS 4300 Database Fundamentals

ITSS 4330 Systems Analysis and Design

ITSS 4351 Foundations of Business Intelligence

ITSS 4360 Network and Information Security

ITSS 4370 Information Technology Infrastructure

ITSS 4382 Applied Artificial Intelligence/Machine Learning

**Guided Electives: 9 semester credit hours**

The BS CIST degree program requires students to focus on a specific concentration to obtain in-depth knowledge in a particular business area.

Students may choose 9 semester credit hours from one of the following Concentrations.

**A. Business Analytics and AI Concentration**

Students who wish to focus on analytical applications on a firm technical foundation can choose the Concentration in Business Analytics and AI, which requires 9 semester credit hours from the following courses:

BUAN 4323 Applied Generative AI for Business

ITSS 4352 Introduction to Web Analytics

ITSS 4353 Business Analytics

ITSS 4354 Advanced Big Data Analytics

ITSS 4355 Data Visualization

ITSS 4383 Machine Learning for Business Analytics

**B. Enterprise Systems Concentration**

Students who wish to focus on the internal, integrated information systems within an organization must take 9 semester credit hours from the following courses:

ACCT 4342 Accounting Information Systems and Financial Reporting

or ACCT 3322 Accounting Transaction Processing and Control

ITSS 4340 Enterprise Resource Planning

ITSS 4343 Supply Chain Information Systems: SAP, AI, and Blockchain Innovations

or ITSS 4344 CRM using Salesforce

**C. IT Sales Engineering Concentration**

Students who wish to focus on sales of technical products and services, must take 9 semester credit hours from the following courses:

ITSS 4340 Enterprise Resource Planning

MKT 3330 Fundamentals of Professional Selling

MKT 3341 AI Within the Sales Technology Landscape

MKT 4331 Digital Prospecting

or MKT 4332 Advanced Professional Sales

**D. IT Innovation and Entrepreneurship Concentration**

Students seeking to learn about developing business startups and/or driving technology innovation inside existing corporations, must take 9 semester credit hours from the following courses:

ENTP 3301 Innovation and Entrepreneurship

ENTP 3320 Start-up Launch I

ENTP 4311 Entrepreneurial Strategy and Business Models

ENTP 4350 Corporate Entrepreneurship

ENTP 4360 Innovation and Creativity

**E. Cybersecurity Management Concentration**

Students who wish to focus on cybersecurity within a firm technical foundation can choose the Concentration in Cybersecurity, which requires 9 semester credit hours from the following courses:

ITSS 4356 Data Governance

ITSS 4357 Digital Forensics and Incident Management

ITSS 4361 Information Technology Cybersecurity

ITSS 4362 Cybersecurity Governance

**F. Information Technology and Systems Concentration**

Any 9 semester credit hours of ITSS upper-division courses, excluding ITSS 4301[8] and ITSS 4V90, that are not part of the BS CIST major preparatory, major core, or major related courses may be used to satisfy the 9 semester credit hours of guided elective credit.

Students in the Jindal Undergraduate Research Scholars program should take BA 3340, BA 3350, and up to 3 SCH of Individual Study based on the major to satisfy JURS program requirements.

These credits may be applied to the Information Technology and Systems concentration.

III. Elective Requirements: 2 semester credit hours

**Free Electives: 2 semester credit hours**

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Indicates a prerequisite class to be completed before enrolling for upper-division classes.

2. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

3. Completion of CS 1436 and/or CS 1337 with a C or better will satisfy ITSS 3311 with an additional 3 semester credit hours of ITSS upper-division courses (excluding ITSS 4301 and ITSS 4V90).

4. Students may use ENGR 2300, TE 2305, MATH 2418, or MATH 2333 as a substitute for OPRE 3333. Students may use CS 3341, ENGR, 3341, SE 3341, STAT 3341, STAT 2332, STAT 3355, STAT 3360, or STAT 4351 as a substitute for OPRE 3360.

5. BPS 4395 can be taken instead of ITSS 4395 only if ITSS 4395 is not available.

6. Students may fulfill the internship requirement with ITSS 4090 or ITSS 4V90 (1-3 semester credit hours). The zero-semester credit hour courses ITSS 4090 is recommended as the most efficient way to satisfy this requirement. If the variable credit internship option is taken, semester credit hours apply as free elective credit towards degree requirements.

7. Students may fulfill the community engagement requirement with BA 4095, BPS 4396, or ENTP 4340 MKT 4360, or OPRE 4391. The zero-semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement. If the variable credit community engagement option is taken, semester credit hours apply as free elective credit towards degree requirements.

8. ACCT 4301 or ITSS 4301 may not be used to satisfy BS CIST degree requirements.

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